B.A PROJECT -Parth Amin

Project Title

Retail Sales Performance Optimization through Business Analysis

1. Project Overview

A retail company operates across multiple regions and sells a variety of products to different customer segments. Recently, management has observed fluctuations in sales and declining profit margins in certain product categories and regions. The company lacks a centralized reporting system to monitor key performance indicators (KPIs) such as sales, profit, discounts, and delivery performance.

This project focuses on analyzing retail sales data to identify performance gaps, monitor business KPIs, and provide data-driven recommendations to improve profitability and operational efficiency.

2. Problem Statement

The retail organization is experiencing:

- Inconsistent sales performance across regions
- Declining profitability due to excessive discounts
- Limited visibility into product and customer performance
- Inefficient reporting processes

Management requires a structured reporting system and actionable insights to support decision-making and improve business performance.

3. Business Objectives

The primary objectives of this project are:

- Monitor sales and profit performance across regions
- Identify high-performing and low-performing products
- Analyze the impact of discounts on profitability
- Improve delivery performance monitoring

- Provide actionable recommendations to increase revenue and profit

4. Stakeholders

Stakeholder	Role	Interest
Sales Manager	Manages sales operations	Improve revenue
Operations Manager	Handles order processing and delivery	Improve efficiency
Finance Manager	Monitors profit and expenses	Increase profitability
Store Manager	Oversees store performance	Optimize product sales
Business Analyst	Analyzes data and reports insights	Support decision-making

5. Project Scope

In Scope

- Sales performance analysis
- Profitability analysis
- Regional performance analysis
- Customer segment analysis
- KPI dashboard development
- Business recommendations

Out of Scope

- Real-time system development
- Customer payment processing
- Inventory procurement automation

BRD SECTION

6. Business Requirements

The system must enable stakeholders to:

- Track total sales and profit
- Monitor performance by region
- Identify top-performing products
- Analyze discount impact on profit
- Monitor delivery performance
- Generate periodic reports

7. Functional Requirements

The system shall:

1. Import retail sales data
2. Calculate total sales and profit
3. Display performance by region
4. Identify top and bottom products
5. Calculate average delivery time
6. Generate summary reports

8. Non-Functional Requirements

Requirement	Description
Performance	Reports should load within 5 seconds
Accuracy	Data calculations must be correct
Usability	Dashboard should be easy to understand
Reliability	Reports should be available when needed
Security	Data access should be restricted

KPI'S

9. Key Performance Indicators (KPIs)

KPI	Description
Total Sales	Total revenue generated
Total Profit	Total profit earned
Profit Margin	Profit percentage
Sales by Region	Sales performance across regions
Top Products	Best-selling products
Average Discount	Average discount given
Delivery Time	Time taken to deliver orders

DATA UNDERSTANDING

10. Dataset Description

The dataset contains transactional data from a retail company, including information about orders, customers, products, sales, profit, discounts, and shipping details.

Total Records:

500+ rows

Key Columns:

Column	Description
Order ID	Unique order identifier
Order Date	Date of order
Ship Date	Date of shipment
Customer Name	Name of customer
Region	Sales region
Category	Product category
Sales	Sales amount
Profit	Profit amount
Discount	Discount percentage

Column	Description
Ship Mode	Delivery method

11. Data Analysis

11.1 Sales Analysis

The total sales performance was analyzed to understand revenue trends across regions and product categories. The analysis helps management identify high-performing areas and focus on revenue growth opportunities.

Key insights to be generated:

- Total Sales
- Sales by Region
- Sales by Category
- Monthly Sales Trend
- Top-Selling Products

Business Purpose:

To monitor revenue performance and identify areas for improvement in sales strategy.

11.2 Profit Analysis

Profit analysis was conducted to evaluate the company's financial performance and identify products or regions contributing to losses.

Key insights to be generated:

- Total Profit
- Profit by Region
- Profit by Category
- Loss-Making Products
- Profit Margin

Business Purpose:

To identify cost inefficiencies and improve profitability.

11.3 Discount Analysis

Discount analysis helps determine how discount policies impact profitability and revenue.

Key insights to be generated:

- Average Discount
- Profit vs Discount Relationship
- High Discount Products
- Discount Impact on Profit

Business Purpose:

To optimize discount strategies and prevent profit loss.

11.4 Regional Performance Analysis

Regional performance analysis evaluates how different geographic regions contribute to overall business performance.

Key insights to be generated:

- Sales by Region
- Profit by Region
- Orders by Region
- Best and Worst Performing Region

Business Purpose:

To support regional planning and resource allocation.

Business Purpose:

To improve logistics efficiency and customer satisfaction.

12. Key Business Insights

Sales Performance

The analysis shows that certain regions contribute significantly more to total sales than others. High-performing regions should be prioritized for marketing and expansion strategies, while low-performing regions require performance improvement initiatives.

Profitability Analysis

Some product categories generate high sales but relatively low profit due to discounts or operational costs. This indicates the need to review pricing and discount policies to maintain profitability.

Sales Trend

Monthly sales trends reveal fluctuations in revenue across different periods. Understanding seasonal patterns can help the company plan inventory, staffing, and promotional campaigns more effectively.

Product Performance

A small group of products contributes a large portion of total sales. These high-performing products should be prioritized for stock availability and promotional activities.

Discount Impact

Higher discount levels are associated with reduced profitability in certain cases. The company should implement controlled discount strategies to prevent revenue loss.

13. Business Recommendations

1. Focus marketing efforts on high-performing regions to maximize revenue growth.
2. Reduce excessive discounts on low-margin products.
3. Monitor monthly sales trends to plan inventory and staffing.
4. Ensure consistent stock availability for top-selling products.
5. Implement a centralized dashboard for real-time performance monitoring.

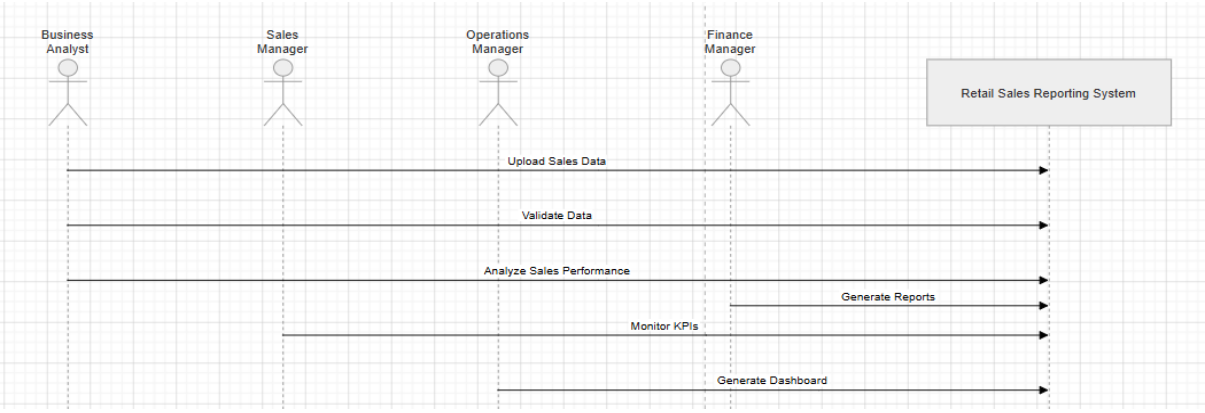
14. Process Flow Description

The process begins with collecting sales data from the retail system. The data is validated to ensure completeness and accuracy. If the data is incomplete, corrections are made before proceeding. Once validated, the data is processed and analyzed to identify trends and performance metrics. The results are then visualized

through dashboards, and business insights are generated. Finally, recommendations are provided to management to support decision-making

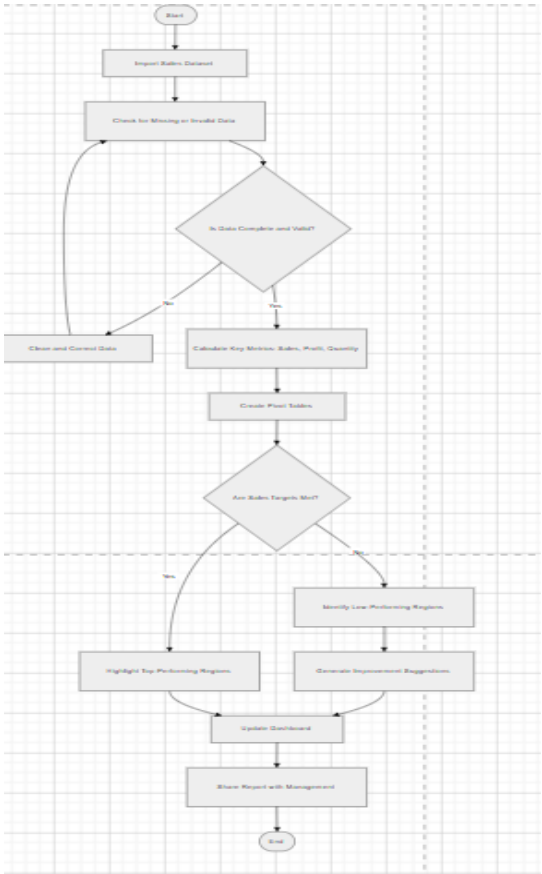
15. Use Case Description

The use case diagram illustrates how different stakeholders interact with the retail sales reporting system. The Business Analyst uploads and validates sales data before performing analysis. The system generates dashboards and reports that allow managers to monitor key performance indicators and make informed business decisions



16. Activity Diagram Description

The activity diagram represents the workflow of analyzing retail sales data. The process begins by loading and validating the dataset to ensure accuracy. If errors are detected, the data is corrected before proceeding. Once validated, the data is analyzed using pivot tables, and dashboards are created to visualize key performance indicators. Business insights are generated from the analysis, and recommendations are provided to support managerial decision-making.



Project Summary

Retail Sales Performance Analysis and Business Insights

This project focuses on analyzing retail sales data to identify performance trends, operational challenges, and business opportunities using structured business analysis techniques. The objective was to transform raw sales data into meaningful insights that support data-driven decision-making and improve overall business performance.

The analysis involved data cleaning, validation, and exploration of key metrics such as total sales, profit, quantity sold, and regional performance. Key Performance Indicators (KPIs) were defined to measure business health, including total revenue, profit margin, top-performing products, and sales by region and category. Pivot tables and dashboards were developed to visualize trends and monitor performance across different states, cities, and product segments.

In addition to data analysis, the project included formal business analysis documentation such as business requirements, process flow diagrams, use case diagrams, and activity diagrams. These artifacts were created to simulate a real-world business environment and demonstrate structured problem-solving, stakeholder-focused thinking, and process improvement skills.

The final outcome of the project provided actionable recommendations to improve sales performance, optimize inventory planning, and identify underperforming regions. This project demonstrates the practical application of business analysis and data analysis techniques to support strategic decision-making in a retail business context.